

Which AI claims in engine performance survive testing?

A measurement rig, and what it found — including about our own numbers

Engineering brief. All results from a pre-registered testbed on a CFM56-7B26 model built entirely from public certification data (TCDS, ICAO EEDB). No proprietary inputs.

Summary. We built a testbed where the true engine condition is known, so a diagnosis can be scored against the answer rather than against another estimate. We then ran the comparison everyone quotes and nobody audits: tuned AI against a tuned classical gas-path stack, on identical data, with hypotheses frozen before the runs. **Most of the advertised advantage did not survive**, including two of our own headline numbers. What did survive is small, specific, and — for an OEM — points at instrumentation rather than software.

1. What failed, and why it matters

A published deep-learning result, replicated

A 2026 Springer paper reports a bidirectional LSTM as the best of fifteen architectures for remaining-life prediction on NASA's C-MAPSS benchmark. We re-implemented it exactly.

- It *does* replicate: 14.56 cycles RMSE against a reported 14.12. This is not a straw man.
- Its architecture ranking rests on gaps of 0.12 to 0.60 cycles. **The run-to-run spread from the random seed alone is 2.94 cycles.** The ranking is noise. The paper reports one run per model.
- The published architecture **fails to train on 5 of 10 runs**, collapsing to predicting zero. The cause is specified in the paper's own tables: a ReLU activation on the output layer, plus a scheduler that cuts the learning rate hundredfold to a floor it cannot recover from.

Takeaway for evaluation: ask any vendor or internal team for the seed spread before you look at the headline. A ranking inside the noise is not a result, and a model that silently fails half the time is not a product.

Two of our own headline numbers, corrected downward

We audited our own claims the same way and both shrank.

- “*AI gives 6× tighter uncertainty intervals.*” False as stated. Our AI interval was statistically calibrated and the classical one was not. Calibrate both the same way — a technique the classical side can adopt for free — and the advantage is **1.34×**. About three quarters of that headline was the calibration method, not the model.
- “*AI predicts remaining life 2.3–4.4× better.*” True only against the linear extrapolation operators field today. Against a competent classical prognostic it is **1.34×** at 90 % of life and a **statistical tie** at 70 %. The advantage is real, late-life, and modest.

Where AI lost outright

Fault isolation between confusable gas-path faults: **31 % versus 31 %**. Identical. The limit is informational, not algorithmic — three cockpit measurements against ten health parameters, with two fault signatures 1.3° apart. We confirmed it four independent ways: halving every sensor noise leaves it at 0.31–0.38; a “virtual sensor” adds *exactly* nothing; sweeping the entire calibrated operating envelope leaves the angle at 1.38°; and no learned method moved it. **No algorithm removes a rank deficiency.**

Also refuted: “physics-informed” learning, twice, when applied to remaining life. And 87 % of mid-life remaining-life uncertainty is *irreducible* — two engines in identical measured condition genuinely fail hundreds of cycles apart. Most prognostic R&D spend is chasing that, not a method gap.

2. What survived

An instrumentation design tool — and it is not AI

The strongest result is classical estimation theory applied through the engine deck: accumulate Fisher information over an engine’s actual flown conditions and you get, per engine and per health direction, a bound on what any diagnosis can possibly know. Validated against the true component state, which only a synthetic-truth testbed permits.

It answers the question an OEM actually owns — and one caveat travels with it, because we found it ourselves and late: the *ranking* claim (that the bound orders the estimator’s true error, $\rho = 0.70$) is real signal but does not clear a physics-free null (0.24 median, $p = 0.085$), because the bound and the estimator share one matrix and a rank test over ten directions cannot separate them. The *acquisition* figures below are separate measurements and are unaffected.

Instrumentation decision	Measured effect
Add station probes (P25/T25/PS3/T3)	HPC efficiency recoverable 45× better
Same probes, confusable isolation	0.15 → 0.92
Halve noise on existing cockpit probes	no effect on the wall
Add a “virtual sensor” / better algorithm	no effect (rank unchanged)

Product-strategy consequence. Any EHM service built on cockpit-only data sits under a hard informational ceiling that no software roadmap removes. The way through is instrumentation — which an OEM controls and a customer does not. This tool prices that decision before the hardware is committed, and it runs on our own decks, where it is stronger than on the public model used here.

The one AI result that held up

Injecting physics into a learner *hurts* remaining-life prediction (refuted twice) but **helps module attribution** — which mechanism is consuming the engine — by a large, significant margin. We ran it with a control arm so the gain could not be attributed to simply adding more inputs.

The reconciliation is mechanical: remaining life is already carried by the EGT-margin signal the model can see, so a physical state estimate adds nothing; *which module* is exactly what that estimate encodes. **“Physics-informed” is neither a free win nor a dead end — the task decides**, and this tells us which of our own EHM products should carry a physics channel and which should not.

A related finding on the wall

The isolation wall is a property of a *single snapshot*. Over a full history, three of five degradation mechanisms become attributable from the shape of the trajectory — including the hot-section mechanism that drives the confusable pair. The governing rule we measured: **trajectory shape separates what has shape**. A wash sawtooth and a break-in knee are readable; a featureless linear ramp is not, however long you watch.

3. What this is, and what it is not

Not: a product, a validated tool for operational use, or evidence about absolute numbers. The engine model is calibrated to public certification data, not to our test-cell data, and the fleet is synthetic. Statistical predictions here are not certified conclusions.

Is: a reusable method for deciding which performance-AI claims to fund and which to reject, with the discipline that makes the answers trustworthy — hypotheses frozen before runs, equal tuning budgets for both sides, negative results reported with the same weight as positive ones, and every figure regenerated from data rather than transcribed.

Proposed next step. Re-run the instrumentation analysis on our own engine decks and our own candidate sensor suites. It requires no AI, no new data collection, and answers a question we already pay to answer by other means.

Every number above traces to a versioned artifact and a pre-registration tag in the study repository; the supporting document runs to 135 pages with the full protocol, the refutations, and the audit trail.